

# PHIL 350: Final

Andrew Bacon

1. Determine whether the following arguments may be proven in the standard system (with axioms A1-A3). If can be, give the proof, if not show that there doesn't exist a proof by appealing to the soundness theorem.

- (a)  $(P \rightarrow (P \rightarrow Q)) \vdash (P \rightarrow Q)$ . (10 points)
- (b)  $\vdash (P \rightarrow Q) \rightarrow Q$ . (10 points)
- (c)  $\vdash (P \rightarrow Q) \rightarrow (P \rightarrow (R \rightarrow Q))$ . (10 points)
- (d)  $\vdash (P \rightarrow (R \rightarrow Q)) \rightarrow (P \rightarrow Q)$ . (10 points)

2. This question is about a new proof system. It is formulated in the language  $\mathcal{L}(\vee, \rightarrow, \neg)$  (so we have  $\vee$  in addition to  $\rightarrow$  and  $\neg$ ). It has axioms:

- B1.  $A \rightarrow (A \vee B)$
- B2.  $(A \rightarrow B) \vee (B \rightarrow A)$
- B3.  $A \rightarrow \neg\neg A$

And two rules of proof

MP. You may infer  $B$  provided  $A$  and  $A \rightarrow B$  have appeared earlier in the proof.

DI. You may infer  $A \vee B$  provided  $A$  has appeared earlier in the proof.

We will use  $\vdash'$  instead of  $\vdash$  to indicate provability in this new system.

- (a) Show that B1 is a (classical) tautology. (5 points)
  - (b) Show that DI is a valid inference:  $A \models A \vee B$ . (5 points).
  - (c) Prove that this system is sound. (You may assume that B1-B3 are tautologies, and DI and MP preserve truth). (15 points).
3. In this question we will show that the above system is not complete. Consider a non-standard notion of truth assignment  $v_0 : Letters \rightarrow \{0, \frac{1}{2}, 1\}$ . A sentence can have the value  $\frac{1}{2}$  as well as 1 or 0. We will call this a *three-valued truth assignment*, and it may be extended to a *three-valued valuation* with the following recursive definition:

- $v(P) = v_0(P)$

- $v(A \vee B) = \max(v(A), v(B))$  (example: if  $v(A) = \frac{1}{2}$  and  $v(B) = 0$  then  $v(A \vee B) = \frac{1}{2}$ ).
- $v(\neg A) = 1 - v(A)$
- $v(A \rightarrow B) = 1$  if  $v(A) \leq v(B)$  and  $v(A \rightarrow B) = 0$  otherwise.  
(PLEASE NOTE: this is different from the usual formula for  $\rightarrow$ ! )

Review the following examples to check you've understood the definition.  
Examples: if  $v(A) = 0$  and  $v(B) = \frac{1}{2}$  then  $v(\neg B) = 1 - \frac{1}{2} = \frac{1}{2}$ .  $v(A \vee B) = \max(0, \frac{1}{2}) = \frac{1}{2}$ . And  $v(A \rightarrow B) = 1$  since  $0 \leq \frac{1}{2}$ .

- Show that the two rules of proof, MP and DI, preserve value 1 in a three-valued valuation: for instance, for DI you must show that if  $v$  is a three-valued valuation and  $v(A) = 1$  then  $v(A \vee B) = 1$ . (5 points)
  - Prove the new axiom system is "sound" for the new notion of valuation. Show that if you can prove  $B$  from  $A_1 \dots A_n$  in this system, and  $v$  is a three-valued valuation such that  $v(A_1) = \dots = v(A_n) = 1$ , then  $v(B) = 1$ . For this question, you may assume that any instance of the axioms, B1, B2 and B3, will have value 1 in any three-valued valuation (this is something I have checked for you). (10 points).
  - Use the soundness with respect to three-valued valuations to show that it is not the case that  $\vdash' (P \vee \neg P)$  (there is no proof, in the new system, of  $P \vee \neg P$  from no premises). (10 points)
  - Explain why this means that the system is not complete. (5 points).
4. In this question you will show that  $\{\wedge, \vee, \neg\}$  are not truth functionally complete for three-valued Kleene logic. Recall that a valuation  $v$  maps sentences to the values 1, 0 or  $\frac{1}{2}$  and  $v(A \wedge B) = \min(v(A), v(B))$ ,  $v(A \vee B) = \max(v(A), v(B))$  and  $v(\neg A) = 1 - v(A)$ .
- Prove by induction that if every propositional letter has value  $\frac{1}{2}$  ( $v_0(P) = \frac{1}{2}$  for every letter  $P$ ) then every sentence defined from  $\wedge, \neg$  and  $\vee$  has value  $\frac{1}{2}$ . (10 points).
  - Define, by giving a truth table, a three-valued connective that cannot be defined from  $\wedge, \neg, \vee$ . You should appeal to your answer to part 1 in explaining why it cannot be defined. (5 points.)